

Design Verification and Validation Plan — Candidate E2

Requirements

ID	Requirement	Source	Metric	Threshold
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R1	Energy density target	Task brief	energy_density_wh_kg	>= 500.94 Wh/kg
R2	4C charging without plating	Task brief	plated	False
R3	Thermal limit under 4C	Task brief	T_max_K	<= 333.15 K

Verification

ID	Method	Result	Status
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R1	Contract-caliber energy calculation	517.830 Wh/kg	PASS
R2	4C lumped thermal simulation with plating model potential 0.000040 V		PASS
R3	4C lumped thermal simulation	330.862 K	PASS

Traceability

- Simulation outputs: "cell/r3_r3h_energy.json", "cell/r3_r3h_4c.json".
- Evaluation entry: round 3 Candidate E2 in "log.jsonl".
- Design parameters: "cell/params_r3h.json".